SafeAid

Presented by:

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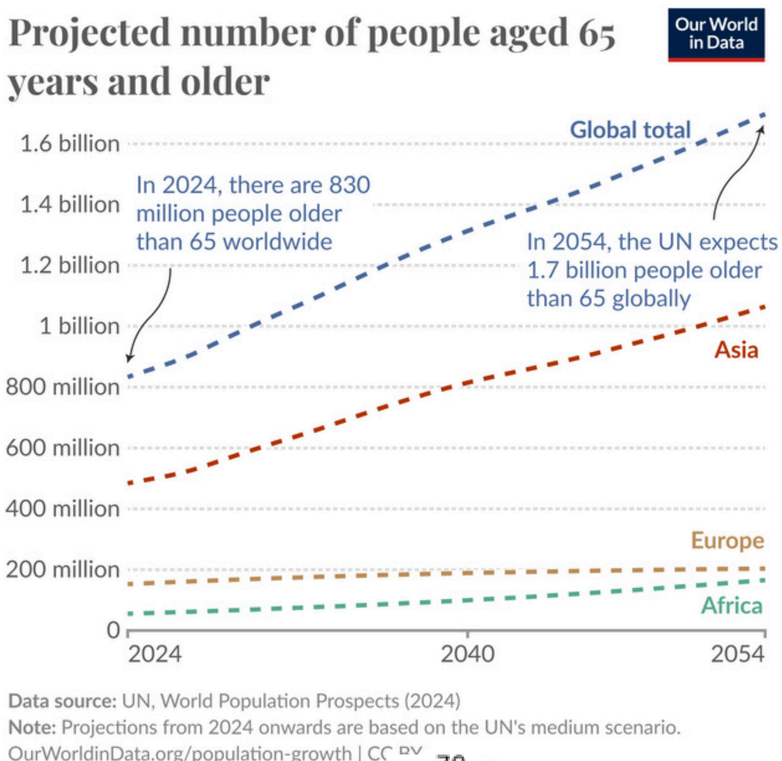
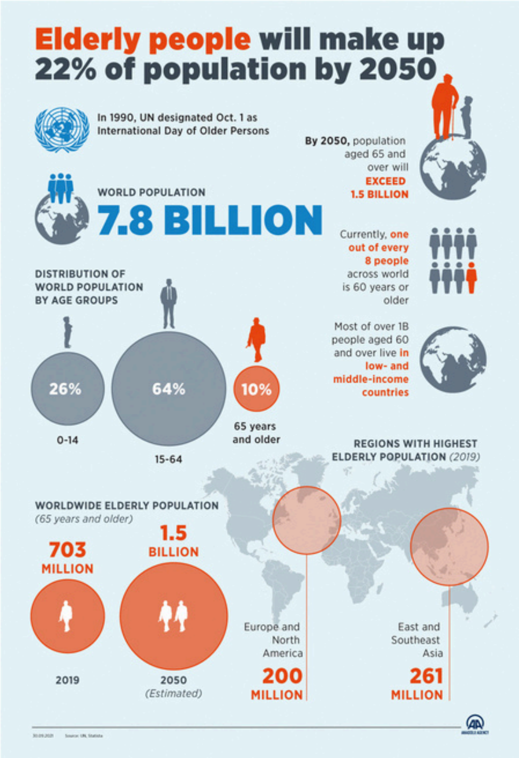
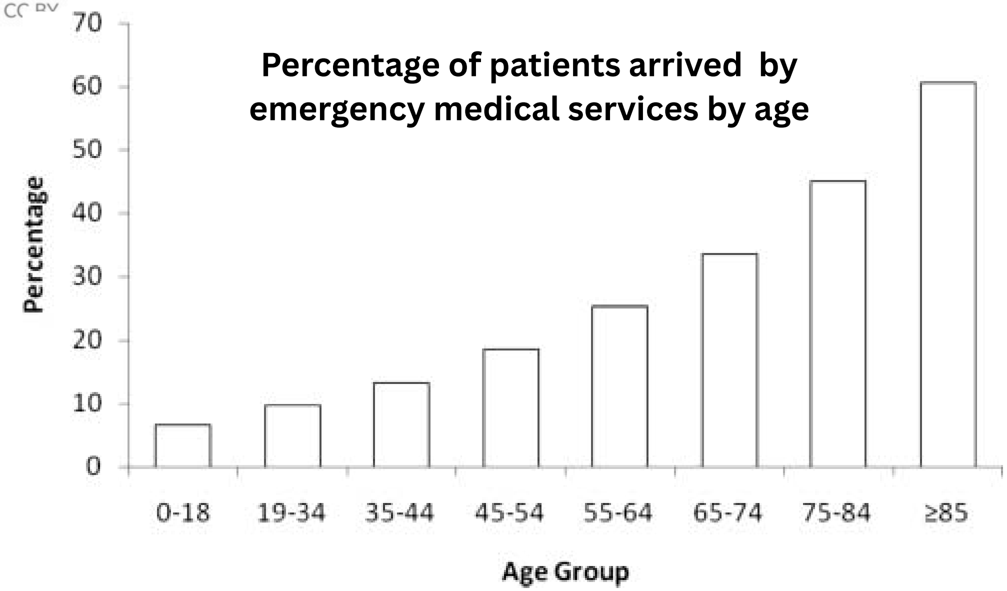
Development Background







Development Background

High Emergency Risk Among People Aged 60 and Above:

- People over the age of 60 have a significantly higher risk of medical emergencies, such as sudden falls and heart-related problems.
- Statistics show that the percentage of emergency incidents increases rapidly after the age of 60, making continuous monitoring essential.
- Many emergencies occur unexpectedly and without warning.

Falls as a Major Cause of Serious Injury

- Falls are one of the most common emergencies among elderly people.
- A fall can lead to fractures, head injuries, or long-term disability.
- In many cases, the injured person is unable to move or call for help after falling.

Delayed Emergency Response

- When an emergency is not detected immediately, the response time increases.
- Delayed medical assistance can result in worsened health outcomes or death.
- Immediate detection and alerting are critical to reducing damage.

Need for Automatic Emergency Detection

- In emergency situations, elderly individuals may be unconscious, confused, or physically unable to press a button or ask for help.
- An automatic system that detects emergencies without user interaction is essential.
- Such a system improves safety and provides reassurance to both users and their families.

Development Objectives

Ensure Rapid Emergency Detection

- To develop a system that can automatically detect emergency situations, such as falls and abnormal heart rates.
- To minimize the time between an emergency event and its detection.

Improve Safety for Elderly Users

- To enhance the personal safety of people aged 60 and above, who are at higher risk of medical emergencies.
- To provide continuous monitoring without interfering with daily activities.

Enable Immediate Emergency Alerts

- To design a system that automatically sends emergency alerts when an abnormal condition is detected.
- To ensure that help can be requested even when the user is unable to respond manually

Provide Real-Time Health Monitoring

- To continuously monitor vital signs, especially heart rate, in real time.
- To detect abnormal patterns that may indicate health risks.

Develop a Low-Cost and Efficient System

- To build a system using affordable hardware components.
- To ensure the system is energy-efficient, reliable, and easy to use.

Team Member Contributions

Khin Moe Moe Latt:

- Soldering and assembly of Arduino sensors
- Circuit design and hardware connections

Saeed Maya Saeed Ibrahim:

- Software development and code implementation
- System logic and sensor data processing

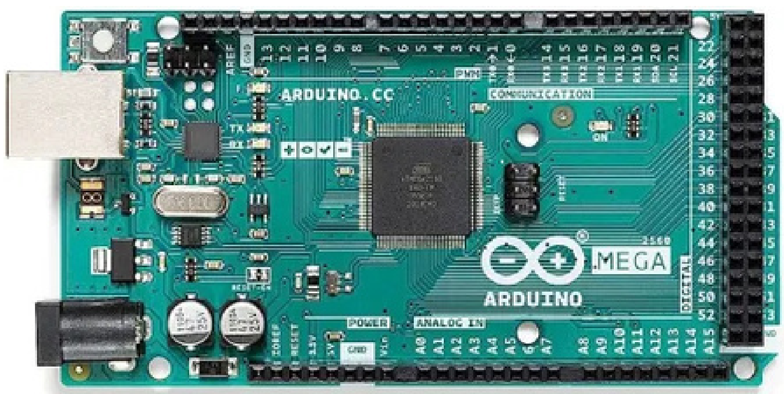
Thon Than Thar:

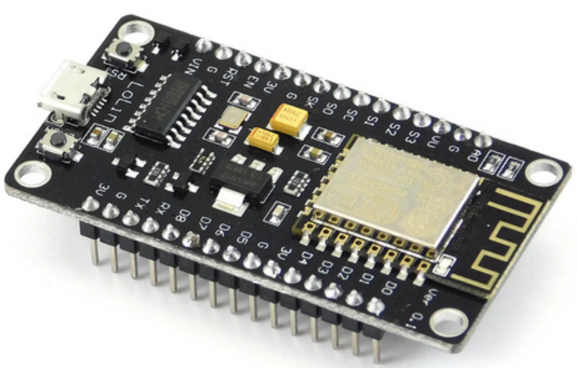
- KakaoTalk application setup and integration
- Emergency notification configuration

Hardware Components

1. Processing & Connectivity

- **Arduino Mega 2560**: Master Controller; manages sensor data and emergency logic.
- **ESP8266 (NodeMCU)**: Communication Gateway; handles Wi-Fi and KakaoTalk API.





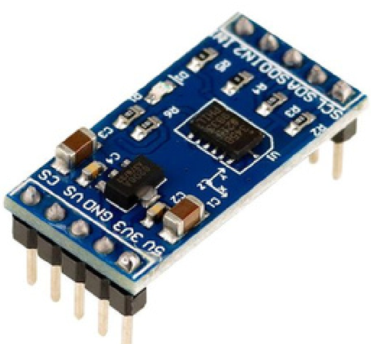
Hardware Components

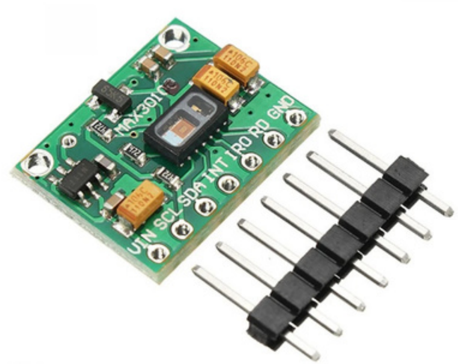
2. Sensing Suite

- **ADXL345 Accelerometer**: Detects physical movement, free-fall, and impact.
- **MAX30102 Heart Rate Sensor**: Optical IR sensor for real-time BPM monitoring.

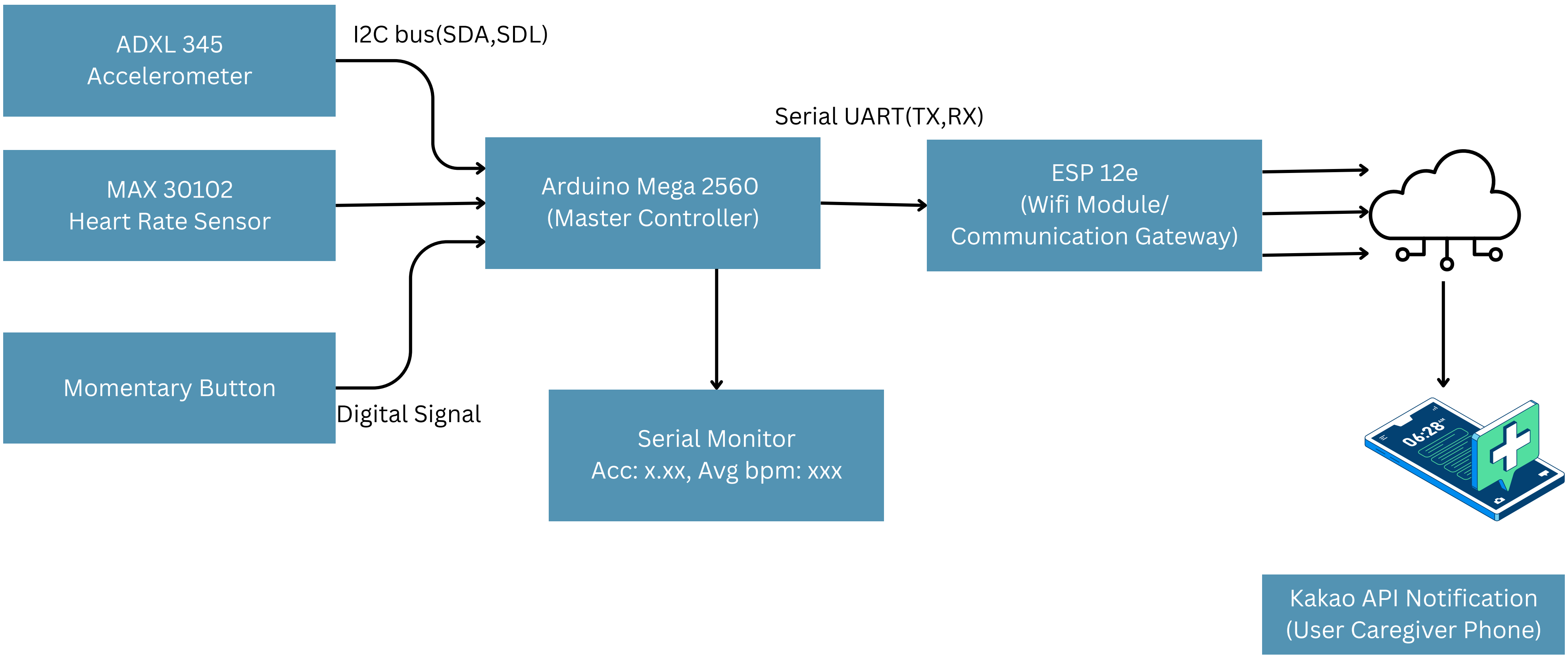
3. User Interface

- **SOS Push Button**: Manual override for instant emergency triggering.

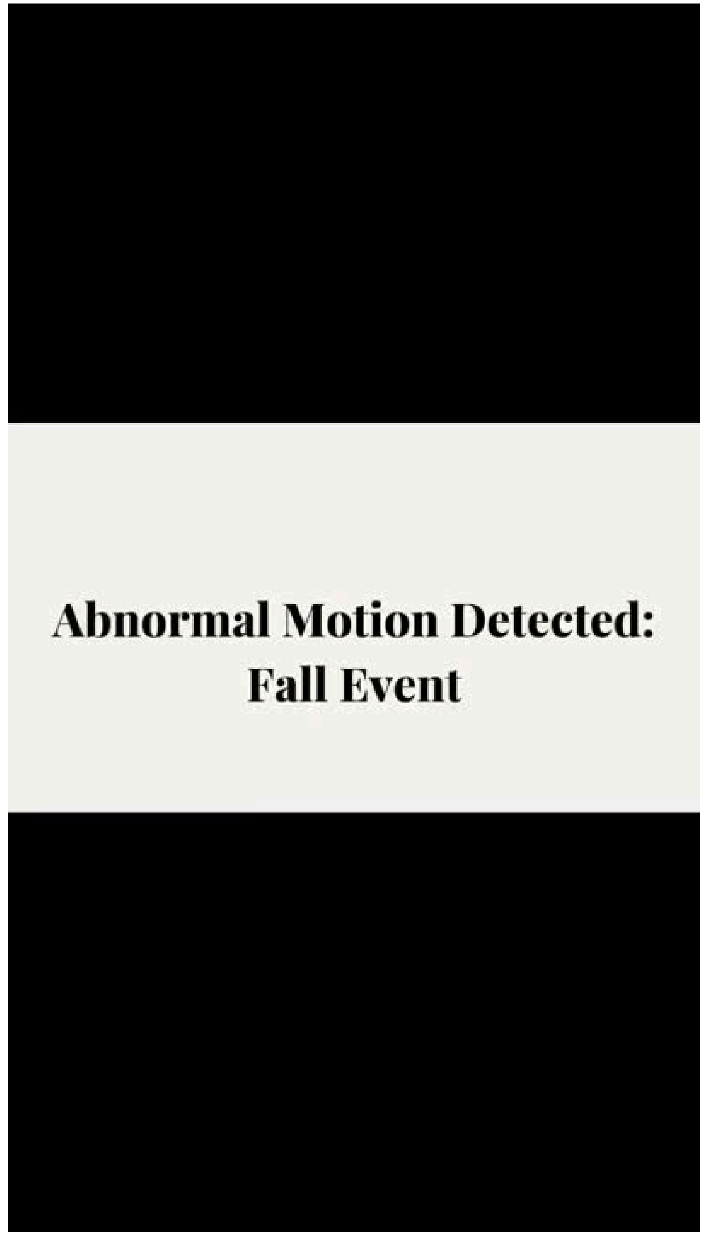




Hardware Components



Demonstration Video



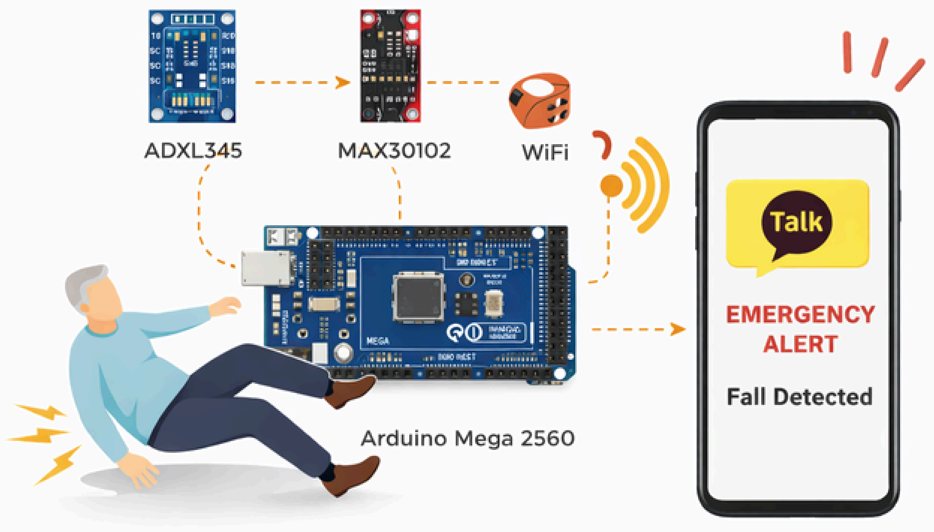
Expected Impact

1. Our System automatically detects emergencies
    - Fall detection
    - Abnormal heart rate
    - SOS button

2. Instant KakaoTalk alert sent to caregivers

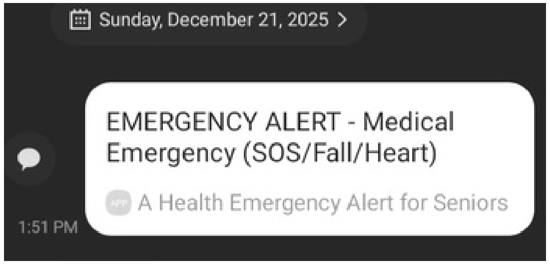
3. Faster response time in emergencies

4. Improves safety, independence, and peace of mind



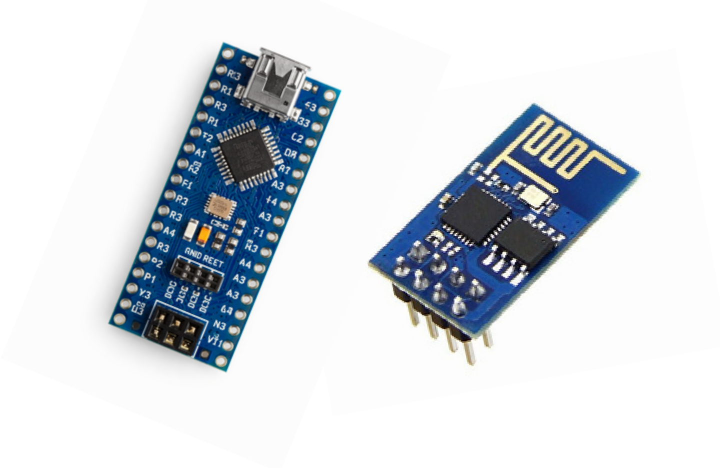
Conclusion

- **Smart health emergency alert system**
- Uses **Arduino**, **health sensors**, and **Wi-Fi**
- Detects : **Falls**, **Abnormal heart rate**, **Manual SOS**
- Sends **real-time emergency alerts** via KakaoTalk
- Shows practical use of **IoT in healthcare**



Future Works

- **Reduce device size** with Arduino Nano R3, ESP8266-01
- Add **3D-printed wearable case**
- Emergency **cancel button** with follow-up alert
- Add health features: **Step counting, Exercise time**
- Add **display or Blynk app** - View health data, Past emergency alerts, Edit emergency contacts
- Send **SMS with user location** for faster rescue



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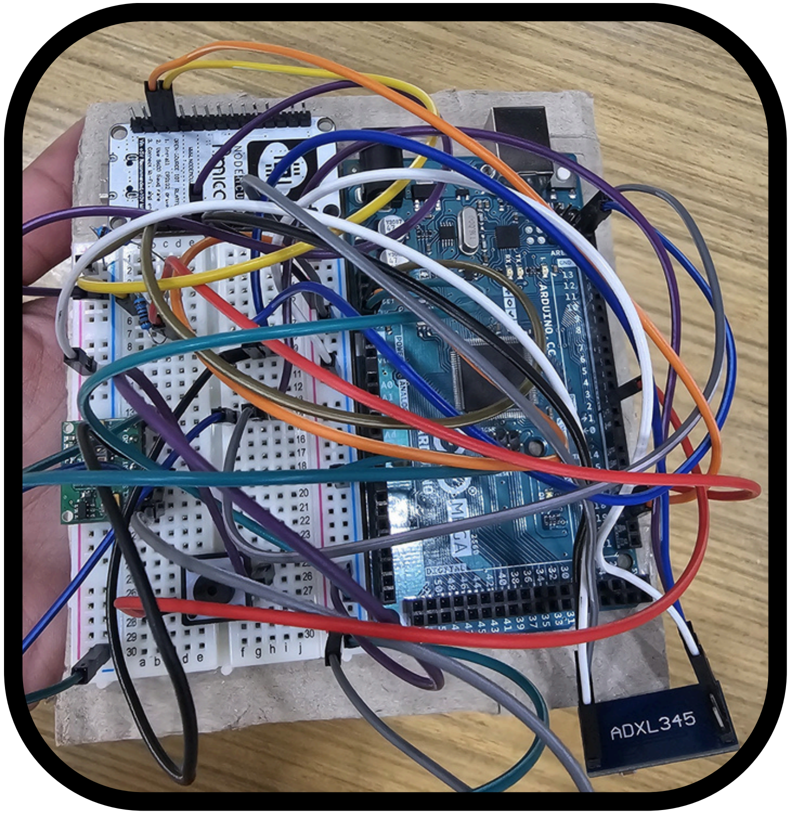
Questions & Answers

🚨Smart Emergency Alert System for Elderly Care



"We care for senior safety"